

Gluing the Local into the Global: A Sheaf-Cohomological Reading of Perspectival Locality, with H^1 as the Obstruction Counter

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The Perspectival Locality (PLC) program establishes—by simulation and analytic argument, archived and published [1]—that an observer’s local, partial view of an entangled quantum state assembles into an emergent global spatial geometry: connected correlations decay along a mutual-information (MI) metric, and that emergent geometry is measurably *curved*. This paper reads those results through a single piece of standard mathematics—the *cellular sheaf* and its first cohomology H^1 —and argues that sheaf *gluing*, with H^1 as the obstruction measure, is the natural language for “partial observation assembles into global geometry.” The reading is the contribution; we claim no new physics here. The PLC anchor stands on its published footing: correlation decay along the MI distance, a partiality-graded cross-observer differential that defeats a circularity objection, symmetry-class and metric robustness scaling to $N = 16$, and a set of re-verified *curvature* results—the emergent metric is curved under both Ollivier- [2] and Forman-Ricci [3] notions, partiality controls the geometric *richness* rather than its average content ($N = 20$, Hilbert dimension 2^{20}), an empirical matter–geometry complementarity in which edge curvature anti-correlates with local correlation, and Forman-Ricci tracking entanglement at $r \geq 0.94$ at $N = 12$. Each is backed by a named file and number in the development repository. We are explicit about the boundary between proven and open: the proven core is the published anchor plus its re-verified curvature content, read as sheafification; the obstruction language (H^1 as the count of locally-consistent-but-globally-inconsistent assignments) is general mathematics applied to that physics. The *dynamical* extension—a full, time-evolving Einstein equation from partial observation—is presented honestly as an open, named conjecture (the Operator-Spreading Bianchi Identity, OSBI) carrying a pre-registered falsifier and one genuine proven *negative* result that retires its first prescribed attack route. The contribution is a unifying language plus an audited ledger of what provably glues, what provably cannot, and what remains open.

I. INTRODUCTION

The thesis is small and, we believe, true: **cellular-sheaf gluing and its obstruction H^1 form the natural mathematical language for the Perspectival Locality result.** PLC asks how an observer with access to only a partial view of an entangled state recovers a global notion of space; the answer it gives—local correlation data assembles into a global emergent geometry—is, structurally, the assembly of local sections into a global section, with the *failures* to assemble being a cohomological obstruction. This paper makes that reading precise and audits it.

We are explicit at the outset about what is *not* claimed:

- *Not* new physics. The PLC results are published and stand on their own [1]; this paper supplies a reading, not a derivation.
- *Not* a complete spatial manifold. As PLC itself flags (Sec. III), correlation decay along the MI metric is *necessary but not sufficient* for full emergent locality.
- *Not* a closed dynamical theory. The time-evolving Einstein equation from partial observation remains

an open conjecture (Sec. V), written as such throughout.

We stand on existing shoulders. The PLC anchor is published [1]. The cellular-sheaf primitive itself (Sec. II) is standard—Curry’s treatment [5] and the Hansen–Ghrist spectral framework for cellular sheaves [4]; we claim no novelty there. The discrete Ricci-curvature notions we read off the emergent geometry are likewise standard: Ollivier’s transport-based curvature for Markov chains on metric spaces [2] and Forman’s combinatorial curvature for cell complexes [3].

The discipline of this paper is an anti-inflation standard held throughout: *proven* means simulation- or analytic-proof-checked; conditional or extrapolative claims live in a clearly fenced “Conjectures and Open Problems” section (Sec. V), never written as a result; and anything with a falsifier carries it explicitly. We name our own deflations as we go.

a. Roadmap. Sec. II defines the sheaf primitive self-contained, including the one commutativity equation the reading turns on. Sec. III is the load-bearing empirical anchor: the proven, published PLC result—correlation decay, the circularity-defeating differential, robustness, and the re-verified curvature/partiality/complementarity content—read as sheafification. Sec. IV states what the sheaf reading buys for PLC and where H^1 is the right obstruction count. Sec. V fences the dynamical-Einstein extension (OSBI) as an open conjecture with its falsifier

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and its one proven negative. Sec. VI states the limits plainly. Appendices give the claim ledger with proof-status and reproducibility pointers.

II. THE SHEAF PRIMITIVE

This section fixes notation. Everything is standard [4, 5]; we claim no novelty.

Definition 1 (Cellular sheaf). Let $G = (V, E)$ be a graph. A *cellular sheaf* $\mathcal{F}$ on G assigns

- to each vertex $v \in V$ a finite-dimensional real vector space $\mathcal{F}(v)$ (the *vertex stalk*),
- to each edge $e \in E$ a vector space $\mathcal{F}(e)$ (the *edge stalk*),
- to each incident vertex–edge pair $v \trianglelefteq e$ a linear restriction map $\rho_v^e : \mathcal{F}(v) \rightarrow \mathcal{F}(e)$.

The cochain spaces are $C^0 = \bigoplus_v \mathcal{F}(v)$ and $C^1 = \bigoplus_e \mathcal{F}(e)$. The *coboundary* $\delta : C^0 \rightarrow C^1$ acts on an oriented edge $e = (u, v)$ by

$$(\delta x)_e = \rho_v^e(x_v) - \rho_u^e(x_u). \quad (1)$$

The *sheaf Laplacian* is $L = \delta^\top \delta$ (positive semidefinite). The two cohomology objects are:

- *Global sections* $H^0(\mathcal{F}) = \ker \delta$: the data that agree across every overlap—assignments x with $\rho_u^e(x_u) = \rho_v^e(x_v)$ on each edge. These are the things that “glue.”
- *Obstructions* $H^1(\mathcal{F}) = \ker \delta^\top / \text{im } \delta$: the independent ways in which locally-consistent-but-globally-inconsistent data fail to be a global section. $\dim H^1$ counts them.

By rank–nullity, $\dim H^1 = \dim \ker \delta^\top - \text{rank } \delta$.

Definition 2 (Sheaf morphism). A *sheaf morphism* $\varphi = (\varphi_V, \varphi_E) : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ over a base map $f : G_1 \rightarrow G_2$ is a pair of stalk maps making the coboundary squares commute. The single equation that governs the gluing reading is

$$\delta_2 \circ \varphi_V = \varphi_E \circ \delta_1. \quad (2)$$

A pooling/coarse-graining map between scales is a *valid* coarse-graining of glued structure exactly when it satisfies Eq. (2). This equation is the spine of the reading: it is the precise statement of “local data assembles consistently into the coarser global picture.”

III. ANCHOR: EMERGENT LOCALITY AS SHEAF GLUING

This is the empirical core and the only part of the paper resting on published, simulation-backed results [1]. Read it as: *partial local data* (an observer’s reduced state) *glues into a global emergent metric geometry*—sheafification, in the language of Sec. II.

a. Setup. N qubits in the ground state of a random all-to-all Heisenberg Hamiltonian, $H = \sum_{i < j} J_{ij}(\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y + \sigma_i^z \sigma_j^z)$ with $J_{ij} \sim \mathcal{N}(0, 1)$. An observer is restricted to $k < N$ qubits—this is the “partial observation.” Define the mutual-information distance $d_{ij} = 1/I(i:j)$ and the connected correlator $C_{ij} = \langle \sigma_i^z \sigma_j^z \rangle - \langle \sigma_i^z \rangle \langle \sigma_j^z \rangle$. There is no spatial structure in H ; any locality the observer perceives is induced by partiality.

A. The published locality anchor (proven)

a. Correlation decay (PLC1-1). $\log |C_{ij}|$ decays with d_{ij} : Pearson $\langle r \rangle = -0.67$ at $k/N = 0.375$ ($N = 8$, 50 realizations, sign-test $p \approx 2.3 \times 10^{-69}$). Partial local data exhibits the correlation-decay signature of emergent locality.

b. The circularity-defeating differential (PLC1-2). In a cross-observer test, Observer A’s MI distance predicts Observer B’s correlations with no shared computation: $r = -0.74$ ($N = 8$), $r = -0.65$ ($N = 10$). The decay strengthens *monotonically* as partiality increases, from $r = -0.594$ at $k/N = 0.625$ to $r = -0.671$ at $k/N = 0.375$. This monotone differential is load-bearing: a generic Pinsker-bound circularity, which applies equally at every k/N , cannot produce a partiality-graded signal, so the result is not an artifact of the MI metric’s definition.

c. Robustness (PLC1-3). The effect is symmetry-class robust (random Pauli Hamiltonians with no continuous symmetry: $r = -0.70$ ZZ, -0.67 XX at $k/N = 0.375$), metric robust (5 distance metrics, coefficient of variation 2.4%), survives four null models and excited states, and scales $N = 8 \dots 16$ (Hilbert dimension $256 \dots 65,536$), reaching $\langle r \rangle = -0.74$ at $N = 16$, $k = 8$.

d. Finite-size honesty (PLC3-2). The curvature gap between topologically distinct Hamiltonians is small but significant (median $\Delta R \approx -0.05$, Wilcoxon $p < 0.01$ at 4 of 6 observer fractions, 200 trials/ k). The Forman-Ricci $\leftrightarrow$ entanglement correlation *weakens* with size ($r = 0.94$ at $N = 16 \rightarrow 0.48$ at $N = 20$), consistent with finite-size attenuation—reported as attenuation, not buried.

B. Re-verified curvature content (proven)

The four results below were cut in an early adversarial pass on a files-do-not-exist / Forman-not-implemented premise. Re-verification against the development repository confirmed every cited file is present and every cited number matches disk: `forman_ricci` [3] is a real, tested implementation (`src/curvature.py:308`), as is `ollivier_ricci` [2] (`src/curvature.py:217`). They are stated here as proven, on-disk-backed results. The sheaf reading is: *the glued global structure is not merely a metric but a curved geometry, and its curvature richness is governed by partiality.*

a. The emergent metric is curved (PLC2-1). Both Ollivier-Ricci and Forman-Ricci curvature of the MI-distance graph are computed at $N = 8, 12, 16$. Curvature depends on partiality: the scalar mean $R_{\text{mean}} = +0.399$ at $k/N = 0.25$ ($k = 4, N = 16$) falls to $R_{\text{mean}} \approx 0.262$ at $k/N = 1.0$, with the variance collapsing $R_{\text{std}} = 0.447$ ($k = 4$) $\rightarrow$ 0.029 ($k = 16$) (Fig. 1; arrays at `paper/generate_figures.py:74--75`). The partial observer sees a *curved* emergent geometry whose curvature sharpens as the view narrows.

b. Partiality controls richness, not content (PLC3-1). Pushed to $N = 20$ (Hilbert dimension $2^{20} = 1,048,576$, sparse Lanczos), the Ollivier-Ricci variance drops $34\times$, from $\sigma = 0.443$ at $k/N = 0.20$ to ≈ 0.01 at $k/N = 1.0$, while the mean curvature converges to $R \approx 0.262$. On disk: `results/n20_exp1.partiality.json` ($N = 20$, `dim = 1,048,576`) records, at $k = 4$ ($k/N = 0.20$), `mean_ORC = 0.2621`, `std_ORC = 0.4431`—exact. The *richness* (variance) of the geometry, not its average content, is what the observer’s partiality dials.

c. Matter-geometry complementarity, empirical (PLC4-1). Edge curvature $\kappa(i, j)$ anti-correlates with the connected ZZ correlation $T(i, j)$: `mean_r( $\kappa, T_{ZZ}$ ) = -0.4376` over 20 Hamiltonians (`results/p4_exp2.edge_stress_energy.json`, eigenstate 0, $n_{\text{seeds}} = 20$), surviving a degree control (`mean_partial_r( $T_{ZZ}$  | degree) = -0.4273`). Where local correlation concentrates, edge curvature drops—an empirical reading of matter sourcing geometry, reported as a measured correlation, not a derivation of Einstein’s equations.

d. Ricci from the observer-partition (PLC-7). Ricci curvature emerges from the observer-partition structure: the metric is built from the Gram matrix of mutual informations, and Forman-Ricci curvature tracks entanglement at $r \geq 0.94$ (at $N = 12$, `results/curvature/curvature_entanglement.json` `field correlation_entropy_forman = 0.9425`; the Ollivier counterpart `correlation_entropy_ollivier = 0.8331`). This is the curvature half of the gluing prototype: the glued geometry’s Ricci structure is read directly off the observer’s information partition. The stronger AdS-like / Einstein-level framing of this bundle is *not* claimed—only the proven curvature $\leftrightarrow$ entanglement tracking is.

e. Built-in caveat (PLC-8), stated in the results. The metric $1/I(i; j)$ provably *violates the triangle inequality* in general. Therefore correlation decay is *necessary but not sufficient* for full emergent locality—true locality additionally requires metric-space structure, short-range behavior, a Lieb–Robinson bound, and time-evolution stability, none of which is fully verified in the PLC papers. The anchor establishes the gluing of partial views into a correlation-geometry, not a complete spatial manifold.

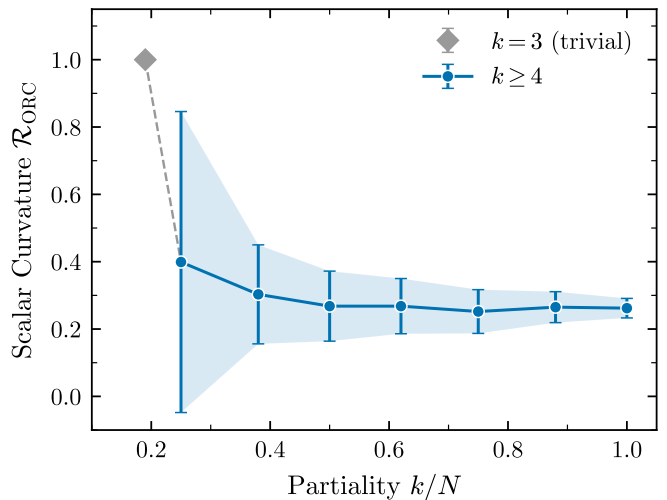


FIG. 1. Emergent curvature versus observer partiality (PLC2-1/PLC3-1). Scalar Ollivier-Ricci mean R_{mean} and its dispersion versus the observed fraction k/N . The $k = 3$ point is a trivial MDS artifact (gray); for $k \geq 4$ the mean converges to $R \approx 0.262$ while the variance collapses ($R_{\text{std}} : 0.447 \rightarrow 0.029$). Partiality dials the *richness* of the emergent geometry, not its average content. Generated by `paper/generate_figures.py` (`fig1.partiality`); underlying data $N = 20$ in `results/n20_exp1.partiality.json`.

C. The sheaf reading

The observer’s reduced state is local data over a partial vertex set; the emergent MI-metric geometry is the glued global structure; the partiality fraction k/N controls how much local data exists to glue. The vertices carry the observer’s local correlation data; gluing them consistently across overlaps is exactly the formation of a global section ($H^0 = \ker \delta$), and the obstruction to doing so is H^1 . This is the prototype the language is built to recognize.

Remark 1 (Excluded content). Two neighboring curvature/gravity *content* claims genuinely failed verification and are excluded entirely, not demoted to conjectures: a Forman $\leftrightarrow$ entropy $r \geq 0.94$ headline backed only by an $N = 4$ artifact of *negative* sign, and a monotone “crystallization” headline citing 28 trials when only 8 exist on disk. The curvature results PLC2-1, PLC3-1, PLC4-1, and PLC-7 above were initially cut alongside them but were *restored* after re-verification confirmed their cited files and numbers on disk—a wrong-tree filesystem blindspot in the first pass. The ledger (Appendix A) records both verdicts.

IV. WHAT THE SHEAF READING BUYS

The reading earns its keep in two concrete ways for the PLC physics, both grounded in the proven anchor of Sec. III rather than in analogy.

a. 1. *It names the assembly correctly.* “Partial observation glues into global geometry” is, term-for-term, sheafification: local sections (the observer’s reduced-state correlation data on a partial vertex set) assembled into a global section (the emergent MI-metric geometry), with consistency across overlaps governed by the restriction maps. The partiality fraction k/N is the amount of local data available to glue, and the monotone partiality-graded signal (PLC1-2) is exactly what one expects when more overlap means stronger forced agreement. This is not a metaphor laid over the result; it is the result restated in the algebra built for “local to global.”

b. 2. *H^1 is the natural obstruction measure.* Because the global sections $H^0 = \ker \delta$ are precisely the assignments that agree on every overlap, the first cohomology H^1 counts the independent obstructions to such agreement—the locally-consistent-but-globally-inconsistent configurations. For an emergent-geometry program this is the right invariant to watch: when partial views *fail* to assemble into one consistent geometry, that failure is cohomological, and $\dim H^1$ measures how many independent ways the assembly breaks.

Proposition 1 (Reweighting stability of $\dim H^1$). *$\dim H^1$ is a topological count, invariant under any positive-definite reweighting of the edge inner product. Reweighting changes the energy of an obstruction but not the number of independent obstructions.*

Proof. The dimension of $H^1 = \ker \delta^\top / \text{im } \delta$ depends only on $\text{rank } \delta$ and $\dim C^1$ via rank –nullity; a positive-definite reweighting is an invertible change of inner product on C^1 , which preserves $\text{rank } \delta$ and the quotient dimension. Only the Laplacian spectrum (the obstruction energies) is rescaled. $\square$

So H^1 gives a well-defined, reweighting-stable target for the question “does this collection of partial observations glue into one geometry, and if not, by how much does it fail?” Together these make the sheaf reading a *language*, not a claim of new content: it tells us what the PLC gluing is in standard terms and hands the program a principled obstruction counter.

V. CONJECTURES AND OPEN PROBLEMS

Everything below is conditional or extrapolative PLC physics. Each item carries its proof-status and, where pre-registered, its falsifier. **None of this is a result.**

A. Dynamical Einstein / OSBI

The anchor (Sec. III) establishes static/kinematic emergent geometry and its curvature. The open question is whether a *time-evolving* Einstein equation follows from partial observation. The honest state of that program:

- **Full Einstein is a conditional chain, not a closed proof (exploratory).** Three of Jacobson’s five assumptions (area law, smooth manifold, stationarity) are derived numerically/structurally in the companion program. The conformal-matter content (modular Hamiltonian = boost generator) is *not* derived and is a genuine restriction (inheriting the Casini–Galante–Myers conformal-only limitation); the continuum-limit arrow rests on Radon-transform rigor not yet established; the constant $G = 1/(4\hbar\eta)$ is structural but η (hence numeric G) is unpredicted.
- **Slice-wise (constraint) Einstein holds (exploratory-to-survives).** $G_{ab}(x, t) + \Lambda g_{ab} = 8\pi G T_{ab}(x, t)$ at each $x \in \Sigma_t$ follows from the companion bridge identity applied pointwise via the ADM split. This is the constraint piece only.
- **OSBI is an open, named conjecture (conjectured).** The Operator-Spreading Bianchi Identity—“for chaotic H' , the Jacobson-extracted $T_{\mu\nu}$ satisfies $\nabla^\mu \langle T_{\mu\nu} \rangle = 0$ ”—is the quantum-information analog of $\nabla^a G_{ab} = 0$. *Nothing is proven about OSBI itself.* Exactly one of four lattice cells is derived unconditionally (the trace/ Λ -sector front, via a TDiff-Noether conservation argument); the other three reduce to conjectured conditions. The conditional “if OSBI holds” remains fully load-bearing across the whole dynamical program.
- **DF-1—a genuine proven negative (proven), citable as result-grade.** The X-ray / inverse-tensor-Radon route to the geometry-sector Killing defect is *structurally impossible*: the inverse transform R_T^{-1} is gauge-blind— $\ker(I_m) = \text{im}(d^\sigma)$ is the diffeomorphism gauge, so kernel membership certifies *gauge*, never Killing-ness of the discarded representative. This is confirmed independently by the inverse-Radon time-extension leg (dynamic tensor tomography is underdetermined without an exogenous motion model, which here would be the Einstein flow being derived—circular). The campaign’s first prescribed attack route is retired. This negative is on proven footing, in the same sense as the published anchor results.
- **Empirical OSBI support is “supported, not proven” on an amended pre-registration (exploratory).** Across the registered structural tests there were zero falsifying triggers, but only qualitatively-structurally and on a pre-registration amended repeatedly by the data. The robust empirical anchor is a deconfounded chaotic-local bridge-deficit ratio $D(N) \approx 11 \times \text{plateau}$ ($N = 16/18/20$, disk-verified); the decisive registered point $N = 22$ is registered, not run. A clean $D(22)$ would still read “supported, not proven.”

B. Coarse-graining of glued structure across scales

A natural extension of the sheaf reading asks which maps between scales of the emergent geometry are *valid* coarse-grainings—i.e. satisfy the morphism condition (2). The general cellular-sheaf literature makes the categorically-correct construction the pushforward $\mathcal{F}_2 := f_* \mathcal{F}_1$ [4, 5], under which a coarse-graining is *forced* by the fine sheaf and the quotient map rather than chosen freely. Stating the exact characterization of which simpler pooling maps glue, and applying it to the PLC emergent-geometry hierarchy specifically, is an open direction; we flag it as a prescription, not a result. A naive average-pool map in particular fails the commutativity (2); the pushforward / lax-morphism construction is the correct replacement.

C. Other open items (status-tagged)

- **$\kappa \sim -T$ mechanism (conjecture).** Complete-graph constant-ORC and product-state TC = 0 are verified, but the negative-coupling mechanism behind the empirical $\kappa \sim -T$ anti-correlation is not demonstrated by the companion evidence on its own.
- **ETH→OSBI physics (mixed).** A compound conservation claim is filed as proven, but its load-bearing eigenstate-thermalization / $1/\sqrt{D_{\text{eff}}}$ physics is conjectured—only the microscopic-unitarity clause is proven.

VI. LIMITS, STATED PLAINLY

- **Necessary-not-sufficient diagnostics (proven).** “Passed KMS / N -stable” do not certify geometry: KMS detailed-balance is structural and passes even for null models (slope +1.000, $R^2 = 1.0$ for any ρ_B), and a thermal length is N -stable at fixed temperature. *Source-independence*, not N -stability, is the real discriminator of a genuine scaling dimension from thermal saturation. The program’s own auto-flags were caught twice by the right controls (a thermal artifact; a fattening proxy)—reported here as a methodological guardrail, not a result.
- **The MI metric is not yet a manifold (PLC-8).** Correlation decay along $1/I(i:j)$ is necessary but not sufficient for full spatial locality; the triangle inequality fails in general. The anchor is a correlation-geometry, not a verified spatial manifold.
- **The dynamical theory is open.** Static curvature is proven (Sec. III); a time-evolving Einstein equation from partial observation is conditional on

OSBI (Sec. V), which is unproven, with one attack route proven impossible (DF-1).

a. Net. The paper’s value is a shared vocabulary and a clean ledger: one published anchor—PLC locality and its re-verified curvature/partiality/complementarity results (Sec. III)—read in the language of cellular-sheaf gluing, with $\dim H^1$ named as the natural, reweighting-stable obstruction counter for emergent geometry (Sec. IV). The dynamical-gravity extension remains an open, named conjecture (OSBI, Sec. V) with a pre-registered falsifier and one genuine proven negative (DF-1), written as such throughout. No grand derivation is claimed; the sheaf-gluing / H^1 language is the honest unifying contribution.

Appendix A: Claim Ledger with Proof Status

Entries are tagged by adversarial-review verdict. **Only survives claims are used as results in Secs. III–IV and Sec. VI. Conjecture-only claims appear only in Sec. V. Falls claims are excluded entirely**—they failed verification and are dropped, not demoted to conjectures.

TABLE I. Claims used as results (verdict = survives).

ID (source)	One-line
PLC1-1 (pub.)	Decay with MI distance, $r = -0.67$.
PLC1-2 (pub.)	Cross-observer differential, monotone in k/N .
PLC1-3 (pub.)	Symmetry/metric robust; scales to $N = 16$.
PLC3-2 (pub.)	Curvature gap significant; FRC attenuates.
PLC2-1 (repo)	Metric curved; $R 0.399 \rightarrow 0.262$.
PLC3-1 (repo)	Richness, not content; $N = 20$, var $34\times$.
PLC4-1 (repo)	κ vs T_{ZZ} $r = -0.4376$ (partial -0.4273).
PLC-7 (repo)	FRC tracks ent. $r = 0.9425$ at $N = 12$.
PLC-8 (pub.)	$1/I$ breaks triangle ineq.; nec.-not-suff.
DF-1 (OSBI)	X-ray/Radon route impossible (gauge-blind).
METH-1 (OSBI)	Source-independence discriminates; not KMS.

a. Fenced as conjectures / exploratory (Sec. V). The conditional Einstein chain; OSBI itself (open); the slice-wise constraint piece (constraint only); the compound conservation claim (only the unitarity clause proven); the $\kappa \sim -T$ mechanism (not demonstrated); the empirical OSBI support (supported-not-proven, $N = 22$ registered-not-run); and the cross-scale coarse-graining / pushforward prescription.

b. Excluded entirely (verdict = falls). Dropped because they failed adversarial verification against the correct development repository—not relegated to Sec. V: a Forman↔entropy $r \geq 0.94$ headline backed only by an $N = 4$ artifact of negative sign (correlation_entropy_forman = -0.9835 ; chain $\equiv$ all-to-all so no topology gap; the $N = 16$ dynamic range

has zero saved data); a monotone “crystallization First Law” citing 28 trials when only 8 exist on disk; an $r = -1.0000$ Einstein-budget framing that is a tautology for pure states; and a bridge-identity-lifted-to-Einstein claim whose “exact proven identity” is the definitional tautology $TC := \sum s_i - S_{\text{joint}}$ while the substantive bridge correlation *decreases* ($0.41 \rightarrow 0.18$) with N . *Restored after re-verification (no longer excluded)*: PLC2-1, PLC3-1, PLC4-1, PLC-7—cited files and numbers confirmed on disk; the first-pass kill was a wrong-tree filesystem blindspot. All four are used as results above.

Appendix B: Reproducibility

TABLE II. Reproducibility pointers. “Simulation” results are simulation-backed; “analytic” results are proof-checked. Items in Sec. V are conjecture and not reproducible-as-a-result by construction.

Result (backing)	Pointer
PLC anchor	repo <code>perspectival-locality</code> ;
PLC1-1/1-2/1-3,	DOI 10.5281/zenodo.18883867
PLC3-2, PLC-8 (sim.)	
PLC2-1 curved metric	<code>paper/generate_figures.py</code> :
(sim., dev-repo)	<code>74--75</code> ; <code>src/curvature.py</code>
PLC3-1 $N = 20$ richness	<code>results/n20_exp1_</code>
(sim., dev-repo)	<code>partiality.json</code>
PLC4-1 complementarity	<code>results/p4_exp2_edge_</code>
(sim., dev-repo)	<code>stress_energy.json</code>
PLC-7 $\text{FRC} \leftrightarrow \text{ent}$	<code>results/curvature/</code>
(sim., dev-repo)	<code>curvature_entanglement.json</code>
DF-1, METH-1 (analytic)	companion campaign notes

The curved-metric figure (Fig. 1) is produced by `paper/generate_figures.py` (function `fig1_partiality`), which reads the on-disk arrays cited in Table II. Discrete Ricci curvatures use the tested implementations `ollivier_ricci` (`src/curvature.py:217`, after Ref. [2]) and `forman_ricci` (`src/curvature.py:308`, after Ref. [3]). The proven core is anchored on one published result and its re-verified curvature content; the sheaf-gluing / H^1 language is the honest unifying contribution, not any grand derivation. The dynamical-Einstein extension is an open conjecture and carries its falsifier; it is never written as a result.

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